



UNIFIED CYBER DEFENSE PLATFORM

SIEM Command Center — Enterprise Threat Detection & Response Console

CompTIA Security+ (SY0-701) Course-End Capstone Project

SIEM & Log Correlation

Vulnerability Mgmt

Network Forensics

Active Directory Security

Automated Hardening

One console for detection, response, and compliance

The platform consolidates SIEM telemetry, vulnerability data, packet capture analysis, Active Directory audits, and automated hardening checks into a single operator console — built to satisfy the CompTIA Security+ SY0-701 course-end project requirements across detection, response, and governance domains.

**14,917**

Security events captured live

**82%**

Current SIEM security posture score

**8**

Monitored enterprise assets

**3**

Core capability domains unified

PROJECT OBJECTIVE

- Correlate logs, alerts & SIGMA rules across every host
- Perform credentialed vulnerability scanning & CVE scoring
- Detect network anomalies via live packet inspection
- Audit Active Directory, GPOs & credential hygiene
- Generate executive-ready PDF & CSV compliance reports

Three-tier architecture, containerized for deployment



Deployment: Docker Compose orchestrates the FastAPI backend, SQLite/Supabase store and the Vite frontend build; Netlify config supports static frontend hosting.

CompTIA Security+ 701 tools mapped to live modules

Category	Tool	Live Module
Network Scanning	Nmap Subnet Engine	Asset Discovery
Vulnerability Mgmt	Tenable Nessus Expert	Vulnerability Management
Penetration Testing	Metasploit (MS17-010)	Threat Detection
Network Analysis	Wireshark / PCAP	Network Intelligence
Log Analysis	System & SSH Logs	Log Analytics
SIEM Concepts	Unified SIEM Engine	Executive Dashboard
Identity & Access	Active Directory (AD DS)	Active Directory Module
Security Policy	Group Policy (GPO)	System Hardening
Cloud / Geo Security	Cloud + GIS Mapping	GIS Breach Map

Six capabilities that set this SOC console apart



AI Security Copilot

Plain-English threat analysis with voice dictation & one-click playbook execution.



3D Attack Globe

Real-time WebGL (Three.js) visualization of active attack vectors and perimeter nodes.



GIS Breach Heatmap

Live IP geolocation tracking (Leaflet) with automated firewall containment.



AD & GPO Auditor

Audits users, security groups, Kerberos tickets and password policy compliance.



SIGMA Correlation Engine

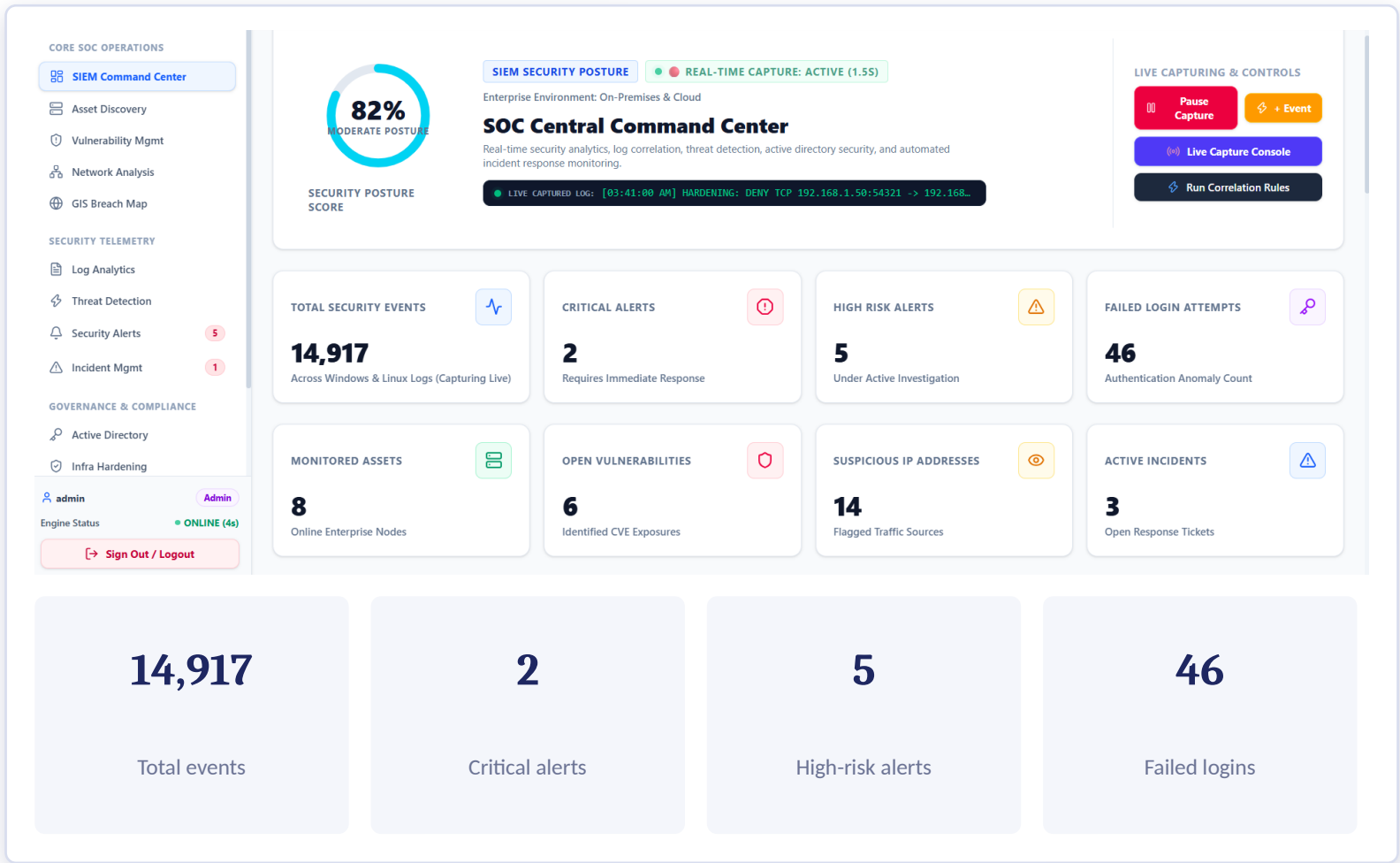
Automated detection rules run continuously against streaming telemetry.



Multichannel Reporting

Web push, Slack/Discord alerts, plus CSV and PDF compliance exports.

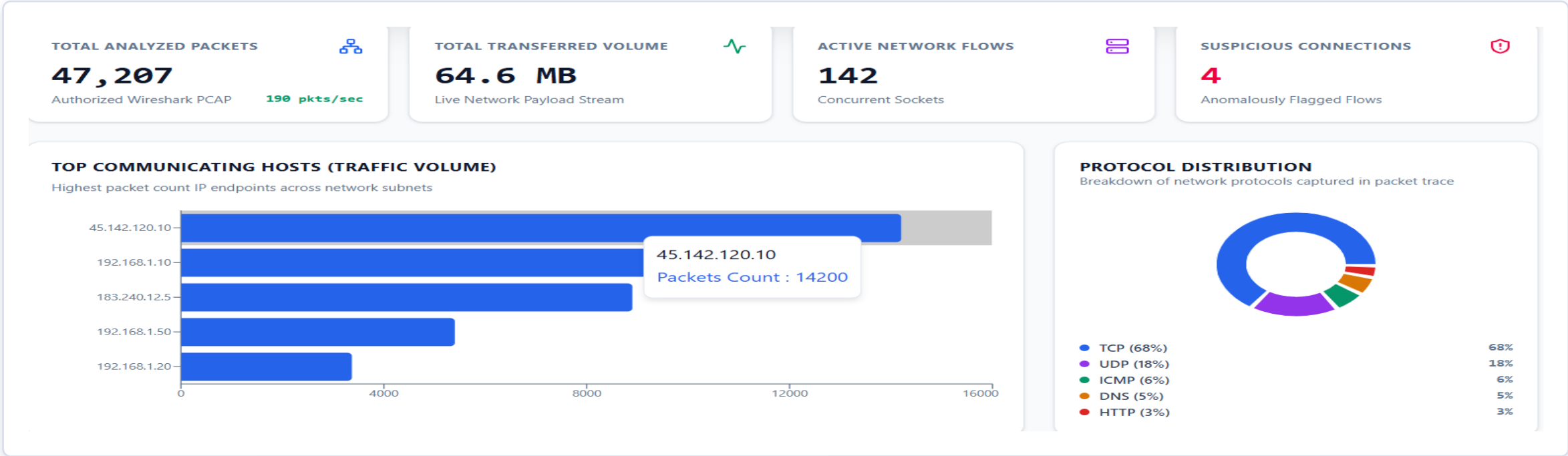
SIEM Command Center: real-time security posture



Live Detection Summary

- 82% moderate security posture score, updated on a 1.5-second real-time capture cycle
- 8 monitored enterprise nodes across on-premises & cloud assets
- 6 open CVE exposures and 14 flagged suspicious IP sources under review
- 3 active incident tickets currently open for SOC response

Network Analysis: packet-level traffic intelligence



 47,207 packets analyzed

Authorized Wireshark PCAP stream at ~190 pkts/sec

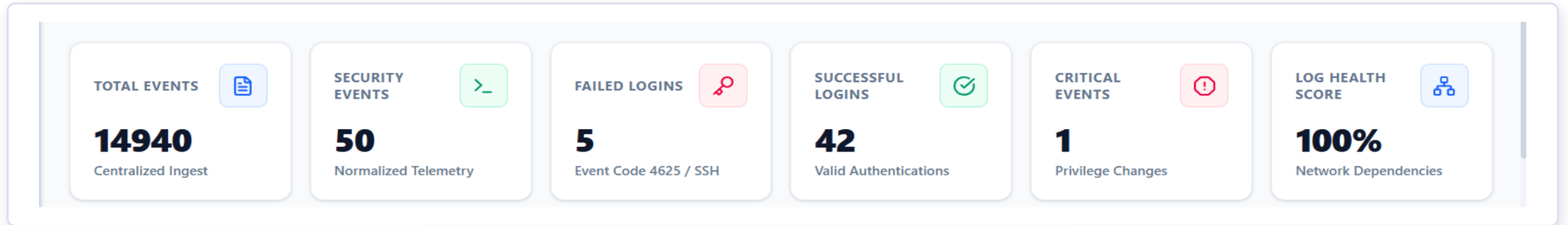
 142 active flows

4 anomalously flagged connections under investigation

 Protocol mix

TCP dominates at 68%, followed by UDP (18%) and ICMP (6%)

Log Analytics: centralized telemetry ingest



What the engine is correlating

14,940	Total events ingested from centralized telemetry
50	Normalized security-relevant events
5 / 42	Failed vs. successful authentications (Event Code 4625 & SSH)
1	Privilege-change critical event flagged for review
100%	Log health score across all network dependencies

Executive reporting & compliance exports

 **EXECUTIVE SECURITY REPORT GENERATION & COMPLIANCE EXPORTER**
Generate formal PDF reports for executive briefings, vulnerability assessments, Active Directory security audits, infrastructure hardening benchmarks, and export raw CSV/Excel telemetry datasets.

**Vulnerability Assessment Report**
Comprehensive report containing all detected CVE vulnerabilities, affected assets, CVSS v3.1 scores, and technical remediation guides.
[Download PDF Report](#)

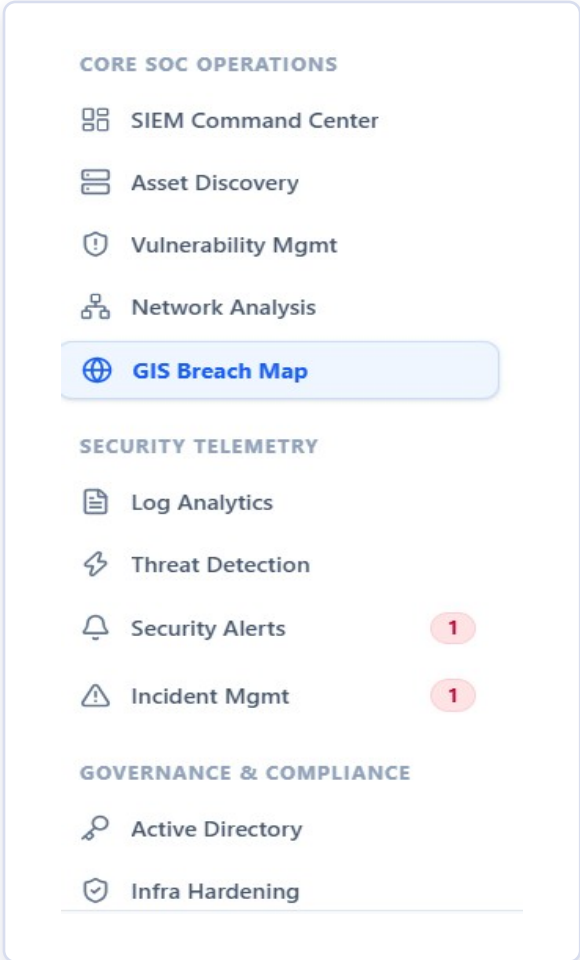
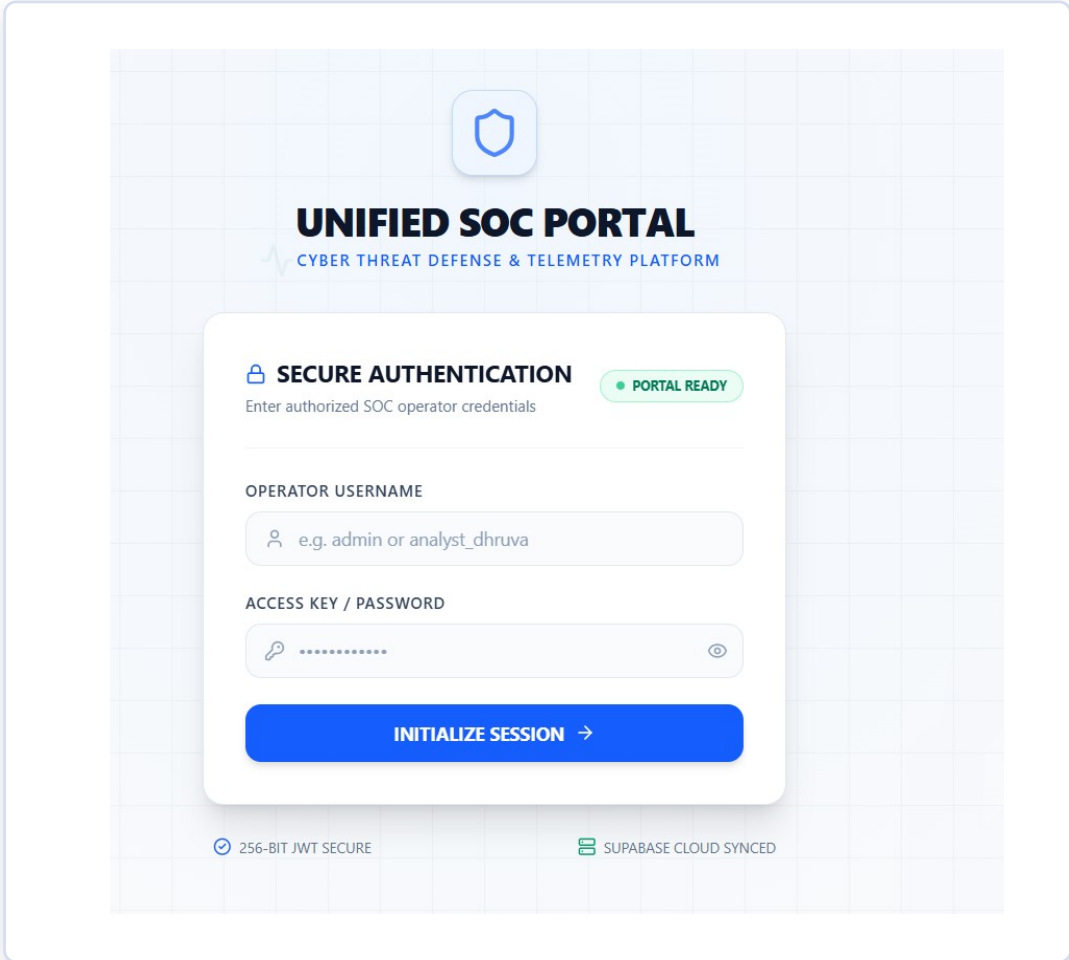
**Executive SIEM Posture Briefing**
High-level executive briefing featuring overall security posture scores, critical alert counts, active threat incidents, and risk trends.
[Download PDF Report](#)

**Infrastructure Hardening Report**
CIS benchmark compliance breakdown across Windows Server and Linux hosts, listing passed controls, failed policies, and warnings.
[Download PDF Report](#)

 **RAW CSV & DATASET EXPORTERS**
Export structured security datasets for external SIEM integration or audit archiving.
[Export Vulnerabilities Dataset \(CSV\)](#) [Export Security Alerts Dataset \(CSV\)](#)

One-click PDF briefings for vulnerability posture, SIEM status and infrastructure hardening — plus raw CSV exports for external SIEM ingestion or audit archiving.

Secure access control & unified navigation



Role-based & auditable by design

- 256-bit JWT authentication with Supabase cloud sync
- Three role tiers: Admin, SOC Manager & Analyst
- 10 modules organized under Core Ops, Telemetry & Compliance
- Session-aware sidebar with live alert & incident badges

Every module traces back to a Security+ course objective

Task 1

Lesson 7

Establishing a Secure Foundation

Kali Linux baseline hardening

Tasks 2–4

Lesson 8

Application & System Security

Nessus scans & Metasploit MS17-010

Tasks 5–7

Lesson 2

Detecting & Correlating Threats

Wireshark PCAP & shell automation

Tasks 8–10

Lesson 10

Identity, Access & Incident Mgmt

AD DS, security groups & GPO policy

Full source code is open on GitHub



Scan to open the repository



kharedhruva-tech / Unified-Cyber-Defense-Platform-SIEM-Command-Center

github.com/kharedhruva-tech/Unified-Cyber-Defense-Platform-SIEM-Command-Center



frontend/

React + Vite + Tailwind operator console



backend/

FastAPI (Python) REST API & SIEM logic



README.md, architecture.md, prd.md

Setup guide, architecture notes & project spec



docker-compose.yml

One-command containerized deployment



A single console, from detection to compliance

The Unified Cyber Defense Platform demonstrates end-to-end SOC operations — network discovery, credentialed vulnerability scanning, live threat correlation, identity governance, and automated executive reporting — unified in one Security+-aligned console.

NEXT STEPS

- Extend SIGMA rule coverage for cloud-native telemetry
- Automate SOAR playbook execution end-to-end
- Add multi-tenant support for MSSP-style deployments

REPOSITORY

Unified-Cyber-Defense-Platform-SIEM-
Command-Center

kharedhruva-tech on GitHub